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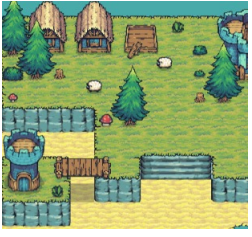
Advanced C++

Initial Team Presentation

Matteo Mauracher, Kian-Alexander Buchholz, Anis Habili

Game Concept

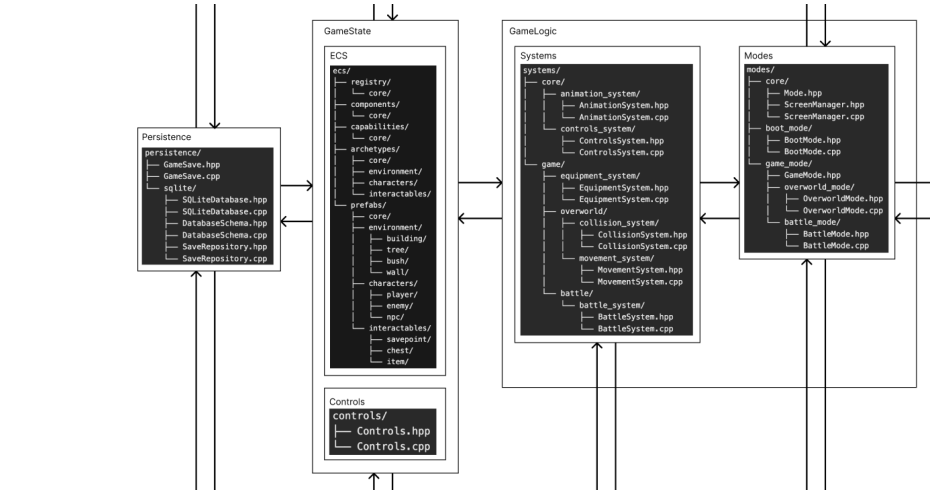
- 2D turn-based RPG.
- Explore the overworld, fight enemies, gain EXP
- Two worlds: the second one is harder, it is unlocked until all first world enemy entities and the mini boss are defeated
- NPCs in both worlds:
  - give hints
  - give boost spells
  - give keys that unlock a secret weapon



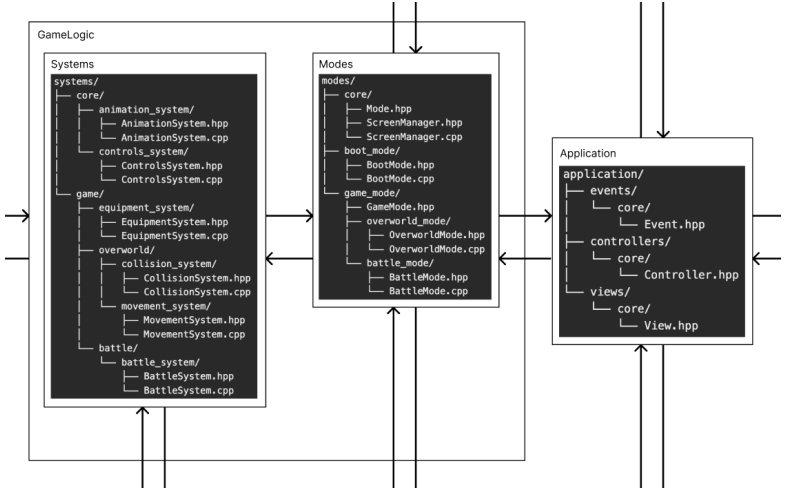
Team and Responsibilities

Goal	Responsible	Points
Overworld + Save points	Anis	3
Combat + Stats	Matteo	4
Main menu + Characters + Resources	Kian	3
Audio	Everyone	1

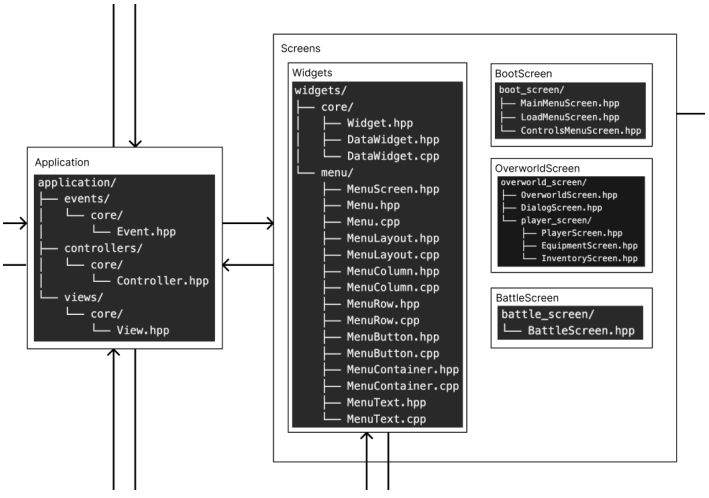
Persistence & Game State



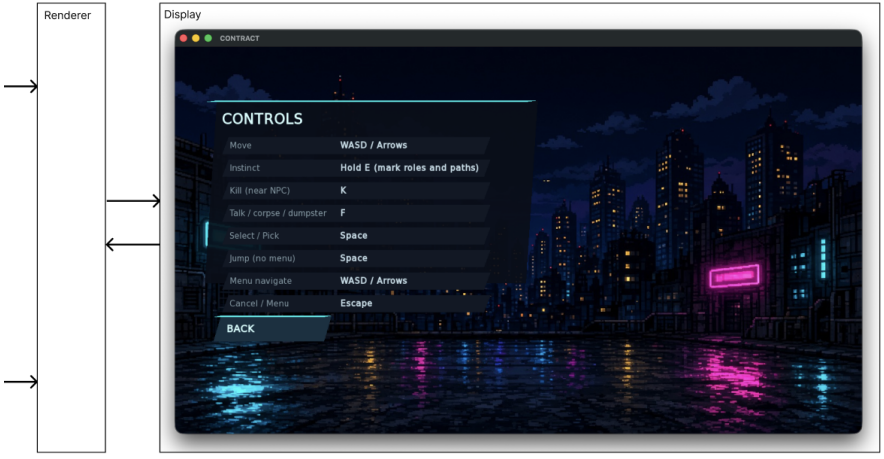
Game Logic


GameLogic
Systems
systems/
├── core/
│ ├── animation_system/
│ │ ├── AnimationSystem.hpp
│ │ └── AnimationSystem.cpp
│ └── controls_system/
│ ├── ControlsSystem.hpp
│ └── ControlsSystem.cpp
└── game/
├── equipment_system/
│ ├── EquipmentSystem.hpp
│ └── EquipmentSystem.cpp
├── overworld/
│ ├── collision_system/
│ │ ├── CollisionSystem.hpp
│ │ └── CollisionSystem.cpp
│ └── movement_system/
│ ├── MovementSystem.hpp
│ └── MovementSystem.cpp
└── battle/
└── battle_system/
├── BattleSystem.hpp
└── BattleSystem.cpp
Modes
modes/
├── core/
│ ├── Mode.hpp
│ ├── ScreenManager.hpp
│ └── ScreenManager.cpp
├── boot_mode/
│ ├── BootMode.hpp
│ └── BootMode.cpp
└── game_mode/
├── GameMode.hpp
├── overworld_mode/
│ ├── OverworldMode.hpp
│ └── OverworldMode.cpp
└── battle_mode/
├── BattleMode.hpp
└── BattleMode.cpp
Application
application/
├── events/
│ └── core/
│ └── Event.hpp
├── controllers/
│ └── core/
│ └── Controller.hpp
└── views/
└── core/
└── View.hpp

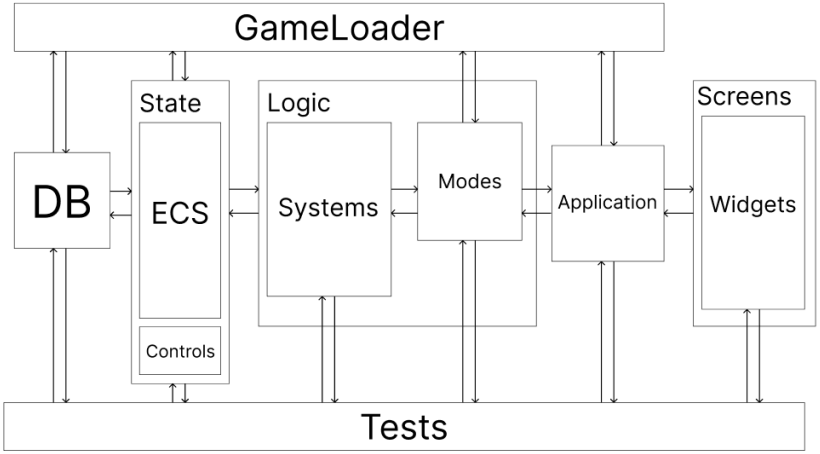
Application Layer & Screens



Assets, Rendering & Display



Game Loader & Tests



Combat

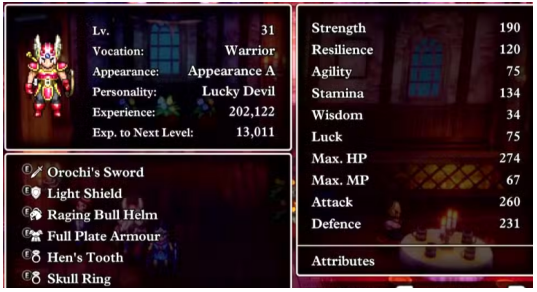
Main Influences: Dragon Quest and EarthBound
Style: Turn-Based Combat Actions; overworld **stealth action sections**
Vibe: Trippy, psychedelic, humorous



Combat Actions: Attack, Skill, Spell, Inventory, Flight

Combat

Vocation / Job-System: influences growth rates
Traditional Stat-Sysem: HP, MP, ATK, DEF, MAG, ...
Learnable Spells / Skills



Dependencies

CMake: using **CMakePresets.json** (crossplatform); **Ninja** as build generator

vcpkg: declarative package management (vcpkg.json)

nlohmann/json: loading the Tiled maps and saving / loading the game progress

SFML: window rendering, sound / music

ImGUI: Real-Time Debug Menus (e.g. Stat Adjustment in Combat)

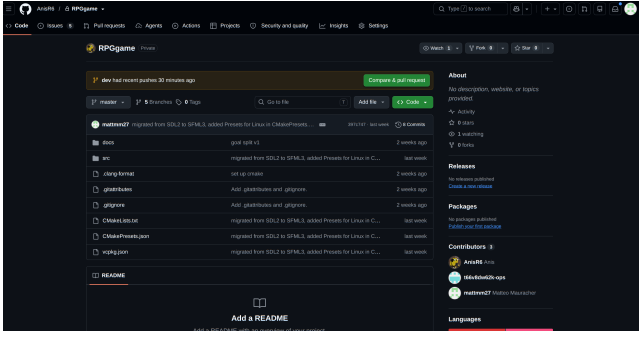


Collaboration

Repository on GitHub

Milestones / Issues

Weekly Meetings on Discord

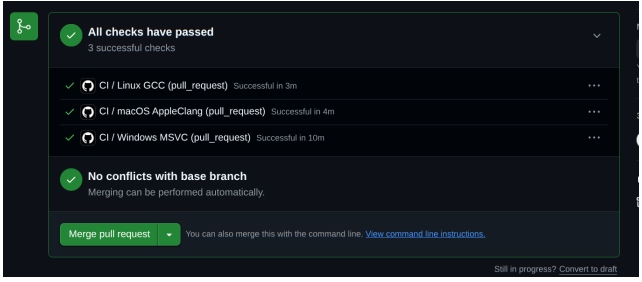


Continuous Integration

GitHub Actions: ci.yml configures, builds and tests crossplatform

Triggered By: Pull Requst or Push to "master" and "dev"

Test Framework: Catch2 (Native Modern C++)



First Milestones

Matteo: Functional, minimal Turn-based Combat

Kian: Architecture + Abstraction Scaffolding, Boot Menu Implementation

Anis: Walkable overworld loaded from Tiled, with player movement and collisions



Thank you for your attention!

Matteo Mauracher, Kian-Alexander Buchholz, Anis Habili